TECHNICAL DATA SHEET

BRSXTG — Xanthan Gum

High-Molecular-Weight Polysaccharide Viscosifier and Suspending Agent for Water-Based Drilling Systems

1. Product Description

BRSXTG is a high-molecular-weight polysaccharide produced by controlled fermentation of Xanthomonas campestris. It is an effective viscosifier and suspending agent for all freshwater- and brine-based drilling systems. It is suitable for slurry TBM, deep foundation, and HDD projects, delivering strong shear-thinning rheology for optimal hole cleaning and cuttings suspension.

The biggest function of BRSXTG is to ensure the stability of drilling fluids used in high salinity areas, especially in marine formations.

2. Main Functions

- Builds viscosity and gel strength in water-based systems
- Enhances cuttings carrying and suspending capacity
- Provides shear-thinning rheology for smooth circulation
- Facilitates efficient solids separation at the surface

3. Product Features

- Cream-colored, free-flowing powder with a flour-like odor
- Highly pseudoplastic — effective at low concentrations
- Soluble in fresh water, brine, and sea water
- Stable over a broad pH range (pH 5.5–8.5, 1% solution)

4. Technical Specifications

Property	Typical Value
Appearance (sensory test)	Off-white or light yellow free-flowing powder or granules
Shearing ratio	7.6
Ash content, %	< 13.0
Pyruvic acid, %	> 5.0

5. Dosage and Mixing Instructions

Recommended dosage: 2–5 kg per cubic meter of water, adjusted to the target viscosity

Mixing: Add the xanthan gum slowly through a venturi hopper and mix with a high-shear mixer for at least 30 minutes. Optimum rheology develops after adequate hydration time

1. Check the water quality (pH and EC) and pretreat if needed.
2. Add the calculated amount of xanthan gum slowly into the mixer.
3. Mix the slurry intensely for at least 30 minutes.
4. Circulate and hydrate the slurry to achieve optimum rheology.

6. Packaging

Small bags: 25 kg per bag

Jumbo bags: Available on request

7. Storage

Keep the product in its original, tightly closed packaging in a cool, dry, and well-ventilated place; protect from moisture, rain, and direct sunlight. Inadequate storage may result in a loss of rheological properties.

8. Supplier Information

Manufacturer / Supplier: Brighter Star Drilling Fluids

Website: www.pcenigeria.com

9. Disclaimer

The technical information and application advice given in this publication are based on the present state of our best scientific and practical knowledge. As the information herein is of a general nature, no assumption can be made as to the product's suitability for any particular use or application, and no warranty as to its accuracy, reliability, or completeness, either expressed or implied, is given other than those required by law. The user is responsible for verifying the suitability of the product for the intended use. In all cases, the actual conditions and results of use shall prevail.

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